A-A-51460
23 June 1986

COMMERCIAL ITEM DESCRIPTION

WHEEL, ABRASIVE, (Aluminum Oxide, Mandrel Mounted)

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-W-37584.

This commercial item description covers three shapes of mandrel mounted abrasive wheels suitable for trimming synthetic resin denture base materials.

Shape 1 - Cone
Shape 2 - Flame
Shape 3 - Bud

Salient characteristics:

The mounted wheels shall be free cutting, self-cleaning and suitable for trimming synthetic resin denture bases during dental prosthetic laboratory procedures. The abrasive shall be white, 54 grit aluminum oxide. The bonding material shall be a porcelain type vitrified binder. The mandrels shall be fabricated from type 416 corrosion resisting steel and shall have a bright finish. The grinding wheels shall be securely mounted on the mandrels with a suitable epoxy or phenolic adhesive. The wheels shall conform to figure 1 for design and dimensions. The wheels shall have a runout of less than 0.005 inch.

Performance. The abrasive wheels shall trim resin denture base material evenly with no deep scratches or coarse rough spots when tested as specified below. The abrasive wheels shall not generate excessive heat, shall have self-cleaning properties and shall not show excessive disintegration when tested as specified below.

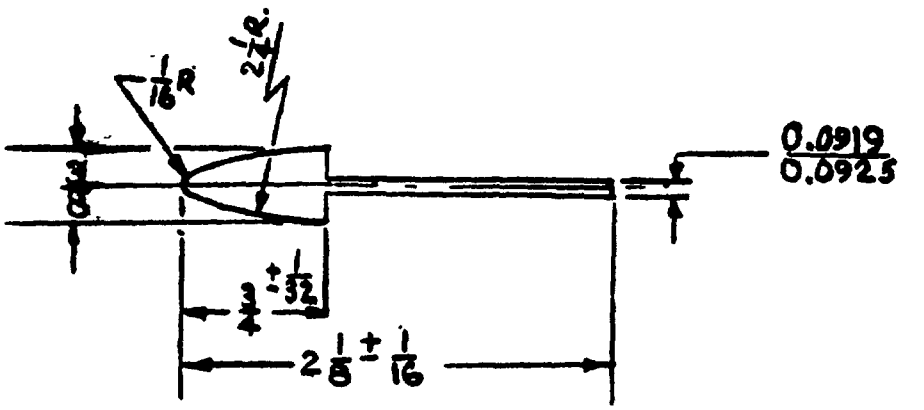
Workmanship. Mandrels shall have no burrs or unspecified sharpness. The mandrel surface shall not be stained or discolored. The abrasive wheels shall be free of dirt, grease, oil, foreign matter or other defects which detract from their appearance or may impair their serviceability.

Unit. Package (PG). One package consisting of six wheels of one shape number only, constitutes one unit.

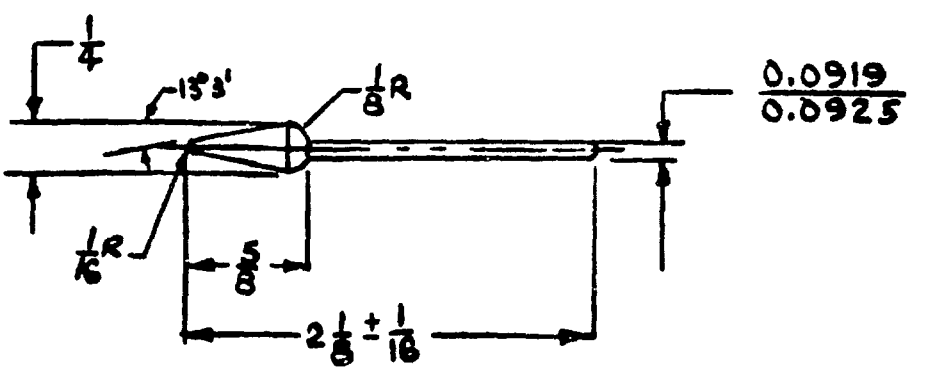
AMSC N/A FSC 6520

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

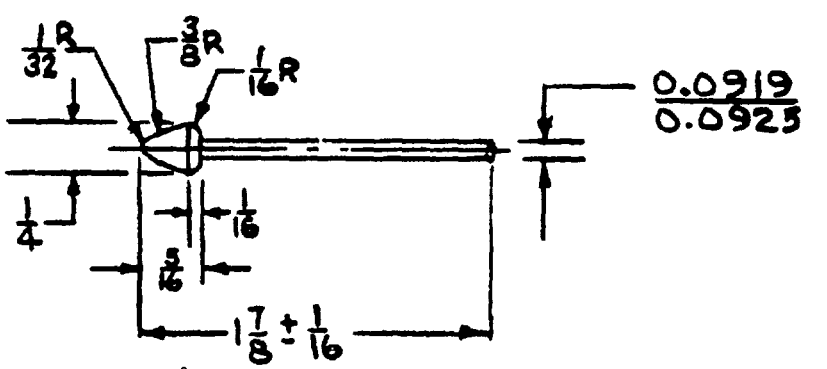
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SHAPE 1-CONE



SHAPE 2-FLAME



SHAPE 3-BUD

Unless otherwise specified, dimensions are in inches and tolerances are as follows:

Fractions:	± 1/64
Angles:	± 1/4°

Figure 1. Wheel, Abrasive

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Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same samples shall be used for the determination of two or more test characteristics.

Performance test. The abrasive wheels shall be used to grind and trim a piece of acrylic resin denture base material approximately 64 millimeters square with minimum thickness of 5 millimeters. The grinding cycle shall be 10 seconds grinding, 20 seconds not grinding for a total accumulated time of 2 minutes. Visual inspection shall be made of the wheel for conformance with the performance requirements.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more test characteristics. The unit of product for inspection purposes shall be one package. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, AQL 2.5, and sampling for tests (when applicable) shall be conducted at inspection level S-1, AQL 1.0.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards, and quality assurance practices.

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The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Numbers are covered by this document:

Stock Number	Item Identification
6520-00-880-6842	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Cone, 3/8 by 3/4 inch, 6s
6520-00-880-6843	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Flame, 1/4 by 5/8 inch, 6s
6520-00-880-6844	WHEEL, ABRASIVE, Aluminum Oxide, Straight Handpiece, Denture Trimming, Bud, 1/4 by 5/16 inch, 6s

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

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Military Interests:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Civil Agency Coordinating Activity
VA-OSS

05450

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

Project No. 6520-1896

Location: NCR

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